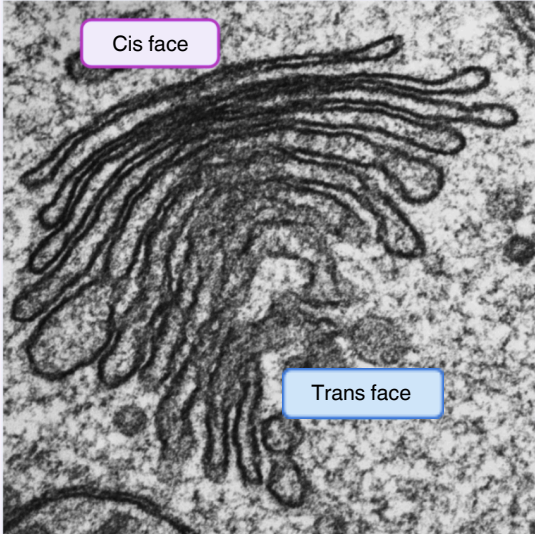
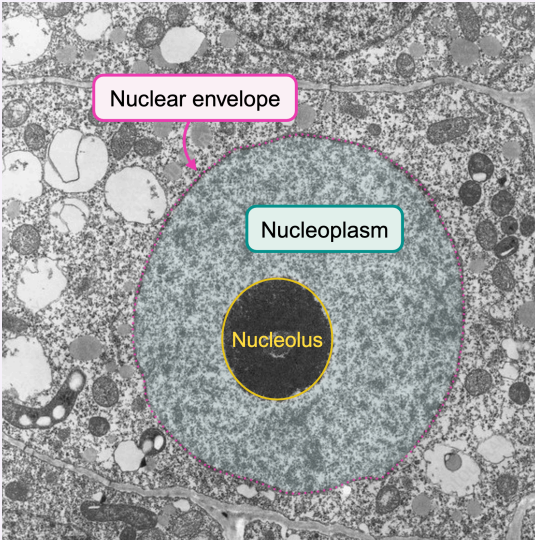
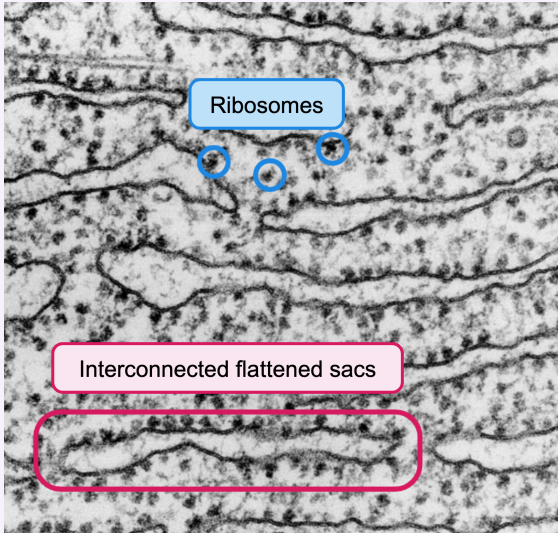
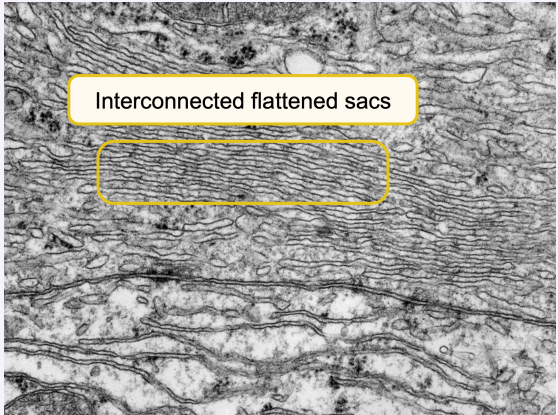


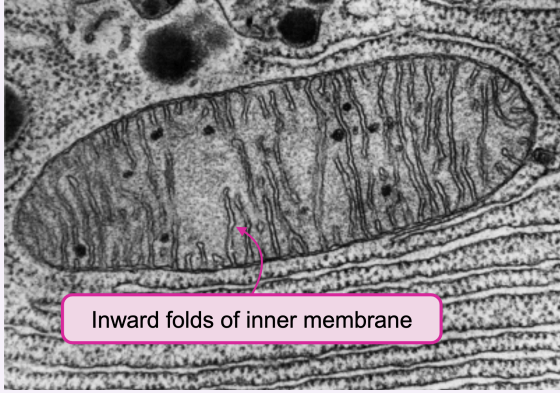
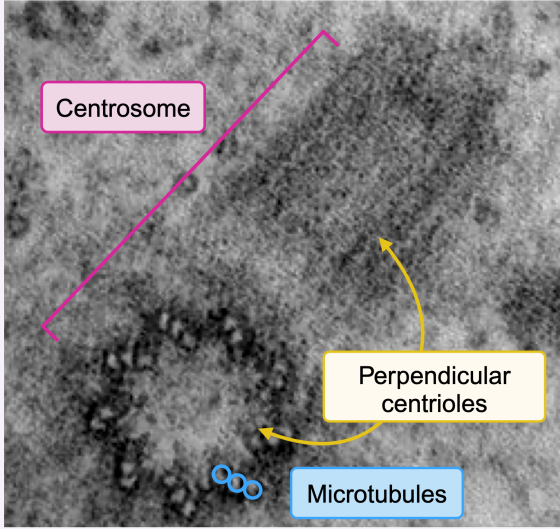
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Structure	Micrograph	Key Features
Golgi apparatus		• Stores, modifies, and exports proteins and lipids destined for secretion from the cell. • Composed of cisternae (flattened sacs) that carry out modification and packaging. • Vesicles originating from the endoplasmic reticulum (ER) approach the Golgi apparatus at the cis face (side closest to the ER). • Vesicles exit the Golgi apparatus from the trans face (side closest to the cell membrane).
Nucleus		• Functions to protect and house DNA, site of DNA replication and transcription (DNA → mRNA). • The nucleoplasm fills the nucleus, similar to cytoplasm in a cell. • A nuclear envelope surrounds the nucleus. • This envelope has openings called nuclear pores, letting molecules move in and out. • Inside the nucleus, the dense nucleolus produces rRNA and combines it with proteins to assemble ribosomes.

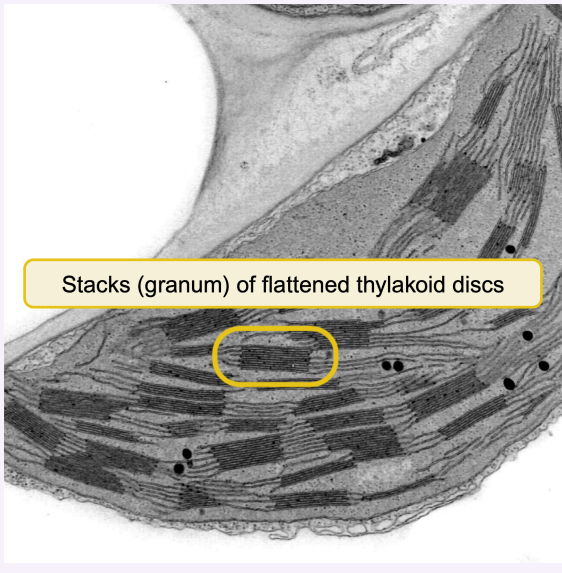
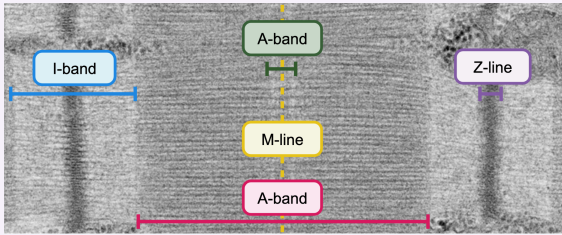
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Structure	Micrograph	Key Features
Rough endoplasmic reticulum		• Continuous with the nuclear envelope and studded with ribosomes. • Proteins synthesized by the embedded ribosomes are sent into the lumen (inside of the rough ER) for modifications (ex. glycosylation) and transport to various locations within and outside of the cell.
Smooth endoplasmic reticulum		• Main function is to synthesize lipids, produce steroid hormones, and detoxify cells.

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Structure	Micrograph	Key Features
Mitochondria		• Powerhouse of the cell, produces ATP through cellular respiration. • Contains its own DNA and reproduces independently via binary fission. • Mitochondrial inheritance is maternal, meaning all mitochondrial DNA in humans originates from the mother.
Centrosome		• Organelle found in animal cells containing a pair of centrioles. • Acts as microtubule organizing centers (MTOCs) during cell division.

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Structure	Micrograph	Key Features
Chloroplast		• Found in plants and some protists that carry out photosynthesis. • Type of plastid, which are double-membraned organelles. • Inside are thylakoids, which conduct light-dependent reactions. These thylakoids stack together to form a structure called a granum.
Sarcomere		• Found in cardiac and skeletal muscle and gives these muscles a striated appearance. • Acts as the functional unit of muscle fibers. • Causes muscles to contract by shortening. • Shortens due to the sliding filament model of muscle contraction.